

MTH-TOP, Topology

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§1 21 September 2026

Roughly speaking, a topological space is a set of *points* together with a notion of which points are “close” to a given one, i.e. a system of neighbourhoods of each point. To specify this, we declare which subsets are *open*.

Definition 1.1 (Topology). Let X be a set and let $\mathcal{P}(X)$ denote its power set, i.e. the set of all subsets of X . A **topology** on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that:

(T1) $\emptyset \in \mathcal{T}$;

(T2) $X \in \mathcal{T}$;

(T3) $\mathcal{T}$ is closed under arbitrary unions: if $\{U_\alpha\}_{\alpha \in A}$ is any collection of elements of $\mathcal{T}$ (the index set A may be finite or infinite), then

$$\bigcup_{\alpha \in A} U_\alpha \in \mathcal{T}; \quad (1.1)$$

(T4) $\mathcal{T}$ is closed under pairwise intersections: if $U, V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$.

The pair $(X, \mathcal{T})$ is called a **topological space**.

Property (T3) allows arbitrary families, while (T4) only concerns pairs. Nevertheless, (T4) implies the corresponding property for finite families, by induction.

Proposition 1.1

Property (T4) is equivalent to:

(T4') for every $n \in \mathbb{N}$ and every $U_1, \dots, U_n \in \mathcal{T}$,

$$\bigcap_{i=1}^n U_i \in \mathcal{T}. \quad (1.2)$$

Proof. (T4') implies (T4) by taking $n = 2$. The converse follows by induction on n , using

$$\bigcap_{i=1}^{n+1} U_i = \left(\bigcap_{i=1}^n U_i \right) \cap U_{n+1}. \quad (1.3)$$

■

When the topology is clear from the context (or not specified), one simply speaks of “the topological space X ”.

Remark 1.1 (Subsets and characteristic functions). There is a natural bijection between subsets of X and functions from X to $\{0, 1\}$:

$$\mathcal{P}(X) \longleftrightarrow \{\psi : X \rightarrow \{0, 1\}\}, \quad A \mapsto \chi_A, \quad \psi \mapsto \psi^{-1}(1), \quad (1.4)$$

where χ_A is the characteristic function of A . One checks easily that these two maps are inverse to each other.

Definition 1.2. Let $(X, \mathcal{T})$ be a topological space. A subset $U \subseteq X$ is **open** if $U \in \mathcal{T}$. A subset $C \subseteq X$ is **closed** if its complement $X \setminus C$ is open.

So closed sets are exactly the complements of open sets. We will use De Morgan's laws: for any family $\{A_\alpha\}_{\alpha \in A}$ of subsets of X ,

$$X \setminus \bigcup_{\alpha} A_{\alpha} = \bigcap_{\alpha} (X \setminus A_{\alpha}), \quad (1.5)$$

$$X \setminus \bigcap_{\alpha} A_{\alpha} = \bigcup_{\alpha} (X \setminus A_{\alpha}). \quad (1.6)$$

The two laws are dual to each other, and proving them is a basic exercise in set theory. Using them, the axioms (T1)–(T4) can be dualized to properties of closed sets.

Proposition 1.2 (Properties of closed sets)

Let X be a topological space.

- (C1) $\emptyset$ and X are closed.
- (C2) An arbitrary intersection of closed sets is closed.
- (C3) A finite union of closed sets is closed.

Proof. (C1) The complement of X in X is $\emptyset$, which is open; the complement of $\emptyset$ is X , which is open. Hence both X and $\emptyset$ are closed. In particular $\emptyset$ and X are both open and closed (“clopen”).

(C2) Let $\{C_\alpha\}$ be a family of closed sets. To check that $\bigcap_{\alpha} C_{\alpha}$ is closed, we must check that its complement is open. By De Morgan,

$$X \setminus \bigcap_{\alpha} C_{\alpha} = \bigcup_{\alpha} (X \setminus C_{\alpha}). \quad (1.7)$$

Each $X \setminus C_{\alpha}$ is open by hypothesis, so the union is open by (T3).

(C3) is proved in the same way, using the other De Morgan law and (T4'). ■

Equivalently, one can *define* a topology by declaring the closed sets, as long as they satisfy (C1)–(C3): the open sets are then the complements of the closed sets.

The following examples are defined on every set X .

Example 1.1 (Trivial/indiscrete topology)

$\mathcal{T}_{\text{triv}} = \{\emptyset, X\}$. The only open sets are $\emptyset$ and X , and these are also the only closed sets.

Example 1.2 (Discrete topology)

$\mathcal{T}_{\text{disc}} = \mathcal{P}(X)$. Every subset is open, and therefore every subset is also closed. Axioms (T1)–(T4) hold trivially since all subsets are allowed.

Example 1.3 (Cofinite topology)

$$\mathcal{T}_{\text{cof}} = \{\emptyset\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}. \quad (1.8)$$

The empty set has to be included by hand in the definition (if X is infinite, $X \setminus \emptyset = X$ is not finite). The open sets are thus the empty set and all subsets with finite complement.

It is not entirely obvious that this is a topology. It is more convenient to check the properties of closed sets. The closed sets of the cofinite topology are exactly X and the finite subsets of X . Then:

- X is closed by definition, and $\emptyset$ is closed because it is finite (it has 0 elements);
- an arbitrary intersection of closed sets is closed: if one of the sets is finite, the intersection is contained in it and hence finite; otherwise all the sets equal X and so does the intersection;
- a union of two closed sets is closed: it is either X (if one of the two is X) or a union of two finite sets, which is finite.

By [Proposition 1.2](#) and the remark above, $\mathcal{T}_{\text{cof}}$ is a topology.

Proposition 1.3

A topological space X is discrete if and only if all its singletons $\{x\}$ are open.

Proof. If X is discrete, every subset is open, in particular every singleton. Conversely, suppose all singletons are open. The empty set is open by (T1), and every non-empty subset $A \subseteq X$ is the union $A = \bigcup_{x \in A} \{x\}$ of open sets, hence open by (T3). ■

Proposition 1.4

If X is finite, then the cofinite topology on X equals the discrete topology on X .

Sketch (exercise). If X is finite, then every subset of X has a finite complement, so every subset is open in $\mathcal{T}_{\text{cof}}$. (If X is infinite, the two topologies differ: a singleton has infinite complement, so it is not cofinite-open.) ■

To construct more interesting topologies, we need the notion of a base. In general, the collection of all open sets is very large and hard to describe explicitly. A base is a smaller collection that can be described in a simple way and that determines the topology.

Definition 1.3 (Base). Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B} \subseteq \mathcal{T}$ of *open* sets is a *base* for $\mathcal{T}$ if every open set of X is a union of elements of $\mathcal{B}$.

It is important that $\mathcal{B}$ consists of open sets, as we'll see now.

Proposition 1.5 (Characterization of open sets via a base)

Let $(X, \mathcal{T})$ be a topological space and let $\mathcal{B} \subseteq \mathcal{T}$ be a collection of open sets. Then $\mathcal{B}$ is a base for $\mathcal{T}$ if and only if for every subset $U \subseteq X$ the following holds:

$$U \in \mathcal{T} \iff \forall x \in U \exists B_x \in \mathcal{B} \text{ such that } x \in B_x \subseteq U. \quad (1.9)$$

Proof. Assume that $\mathcal{B}$ is a base. Suppose that for every $x \in U$ there is $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U$. Then $U = \bigcup_{x \in U} B_x$. Each B_x is open, so U is a union of open sets and is therefore open.

Conversely, suppose U is open. Since $\mathcal{B}$ is a base, $U = \bigcup_{i \in I} B_i$ for some $B_i \in \mathcal{B}$. If $x \in U$, then $x \in B_i$ for some i , and $B_i \subseteq U$; so $B_x := B_i$ works.

The converse (the property implies that $\mathcal{B}$ is a base) follows by the same argument. For every open U and every $x \in U$, there is $B_x \in \mathcal{B}$ with $x \in B_x \subseteq U$. Then $U = \bigcup_{x \in U} B_x$, is an union of the basic open sets. Every open set is a union of elements of $\mathcal{B}$, so $\mathcal{B}$ is a base. ■

Example 1.4

For every set X , the collection of all singletons $\{\{x\} : x \in X\}$ is a base for the discrete topology: singletons are open in the discrete topology, and every subset is a union of singletons (Proposition 1.3).

The following theorem tells us when a collection of subsets of a set (with no topology given yet) is a base for some topology.

Theorem 1.6

Let X be a set and let $\mathcal{B} \subseteq \mathcal{P}(X)$ be a collection of subsets of X . Then $\mathcal{B}$ is a base for some topology on X if and only if the following two properties hold:

- (B1) $X = \bigcup_{B \in \mathcal{B}} B$;
- (B2) for every $B_1, B_2 \in \mathcal{B}$ and every $x \in B_1 \cap B_2$, there exists $B_3 \in \mathcal{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$.

Moreover, in this case the topology is unique: there is exactly one topology on X having $\mathcal{B}$ as a base.

This is very useful: we do not need to describe the (large) collection of all open sets, only a simple collection $\mathcal{B}$ satisfying (B1) and (B2), and this determines a unique topology.

Proof. ($\Rightarrow$) Suppose $\mathcal{B}$ is a base for a topology $\mathcal{T}$. Then X is open, hence a union of elements of $\mathcal{B}$; since every element of $\mathcal{B}$ is a subset of X , this union is X , which gives (B1). If $B_1, B_2 \in \mathcal{B}$, they are open, so $B_1 \cap B_2$ is open, and for $x \in B_1 \cap B_2$ Proposition 1.5 gives $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$, which is (B2).

($\Leftarrow$) Assume (B1) and (B2). Define

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{j \in J} B_j : J \text{ arbitrary index set, } B_j \in \mathcal{B} \right\}, \quad (1.10)$$

that is, the collection of all possible unions of elements of $\mathcal{B}$. We must show that $\mathcal{T}_{\mathcal{B}}$ is a topology.

First, we record a description of $\mathcal{T}_{\mathcal{B}}$ analogous to Proposition 1.5. Note that at this point X is just a set, so we cannot use that proposition; we prove directly that for $U \subseteq X$,

$$U \in \mathcal{T}_{\mathcal{B}} \iff \forall x \in U \exists B_x \in \mathcal{B} : x \in B_x \subseteq U. \quad (1.11)$$

If $U \in \mathcal{T}_{\mathcal{B}}$, then U is a union of elements of $\mathcal{B}$, so each $x \in U$ lies in one of them, and that one is contained in U . Conversely, if the right-hand side holds, then $U = \bigcup_{x \in U} B_x \in \mathcal{T}_{\mathcal{B}}$.

Now we check the axioms.

- (T1): $\emptyset$ is the empty union, so $\emptyset \in \mathcal{T}_{\mathcal{B}}$.
- (T2): $X \in \mathcal{T}_{\mathcal{B}}$ by (B1).
- (T3): a union of unions of elements of $\mathcal{B}$ is again a union of elements of $\mathcal{B}$.
- (T4): let $U, V \in \mathcal{T}_{\mathcal{B}}$ and $x \in U \cap V$. By (1.11) there are $B, B' \in \mathcal{B}$ with $x \in B \subseteq U$ and $x \in B' \subseteq V$. By (B2) there is $B'' \in \mathcal{B}$ with

$$x \in B'' \subseteq B \cap B' \subseteq U \cap V. \quad (1.12)$$

Hence $U \cap V$ satisfies the right-hand side of (1.11), so $U \cap V \in \mathcal{T}_{\mathcal{B}}$.

So $\mathcal{T}_{\mathcal{B}}$ is a topology. Every element of $\mathcal{B}$ belongs to $\mathcal{T}_{\mathcal{B}}$ (a union with a single term), and by definition every element of $\mathcal{T}_{\mathcal{B}}$ is a union of elements of $\mathcal{B}$; thus $\mathcal{B}$ is a base for $\mathcal{T}_{\mathcal{B}}$.

Uniqueness. Let $\mathcal{T}'$ be any topology on X for which $\mathcal{B}$ is a base. Every $U \in \mathcal{T}'$ is a union of elements of $\mathcal{B}$, so $U \in \mathcal{T}_{\mathcal{B}}$. Conversely, every element of $\mathcal{T}_{\mathcal{B}}$ is a union of elements of $\mathcal{B} \subseteq \mathcal{T}'$, hence belongs to $\mathcal{T}'$ by (T3). Thus $\mathcal{T}' = \mathcal{T}_{\mathcal{B}}$. ■

Now, we'll see some examples of topologies on $\mathbb{R}$.

Example 1.5 (The Euclidean topology). Let

$$\mathcal{B}_E = \{(a, b) : a, b \in \mathbb{R}, a < b\} \quad (1.13)$$

be the collection of all open bounded intervals in $\mathbb{R}$. (Here (a, b) denotes the open interval $\{x \in \mathbb{R} : a < x < b\}$; the notation $]a, b[$ is sometimes used to avoid confusion with ordered pairs.)

We check the hypotheses of [Theorem 1.6](#). Property (B1) holds because $\mathbb{R} = \bigcup_{n \in \mathbb{N}} (-n, n)$. For (B2), let $(a, b), (c, d) \in \mathcal{B}_E$ and $x \in (a, b) \cap (c, d)$. Then

$$x \in (\max\{a, c\}, \min\{b, d\}) = (a, b) \cap (c, d), \quad (1.14)$$

and the right-hand side is again an open bounded interval (it is non-empty, since it contains x).

Hence $\mathcal{B}_E$ is a base for a unique topology on $\mathbb{R}$, called the *Euclidean topology* (also the *standard topology*). This is the topology used in calculus and analysis, and it is the default topology on $\mathbb{R}$: when we speak of $\mathbb{R}$ as a topological space without specifying the topology, we mean this one. The elements of $\mathcal{B}_E$ are called *basic open sets*.

- Open intervals (a, b) are open, being basic open sets.
- The rays $(a, +\infty)$ and $(-\infty, b)$ are open, since

$$(a, +\infty) = \bigcup_{n \in \mathbb{N}} (a, a + n), \quad (-\infty, b) = \bigcup_{n \in \mathbb{N}} (b - n, b), \quad (1.15)$$

are unions of basic open sets.

- Singletons are closed: the complement of $\{a\}$ is

$$\mathbb{R} \setminus \{a\} = (-\infty, a) \cup (a, +\infty), \quad (1.16)$$

a union of two open sets.

- Closed intervals $[a, b]$ are closed, since $\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, +\infty)$ is open.
- Half-open intervals such as $[a, b)$ are in general neither open nor closed in the Euclidean topology.

Example 1.6 (The Sorgenfrey line). Now consider a different collection:

$$\mathcal{B}_S = \{[a, b) : a, b \in \mathbb{R}, a < b\}. \quad (1.17)$$

Proposition 1.7

$\mathcal{B}_S$ is a base for a topology on $\mathbb{R}$, called the *Sorgenfrey topology* (the *Sorgenfrey line*, or the topology of right half-open intervals), denoted $\mathbb{R}_S$.

Proof. (B1) holds because $\mathbb{R} = \bigcup_{n \in \mathbb{N}} [-n, n]$. For (B2), the intersection of two such intervals is $[a, b] \cap [c, d] = [\max\{a, c\}, \min\{b, d\}]$; if it contains a point x , it is itself a non-empty interval of the same form, so it belongs to $\mathcal{B}_S$, and we can take B_3 equal to the intersection. ■

Comparison with the Euclidean topology. Let $a < b$. For every c with $a < c < b$, the interval $[c, b]$ is a basic open set of the Sorgenfrey topology, and

$$(a, b) = \bigcup_{a < c < b} [c, b]. \quad (1.18)$$

Therefore every basic open set of the Euclidean topology is open in the Sorgenfrey line, and consequently

$$\mathcal{T}_E \subseteq \mathcal{T}_S. \quad (1.19)$$

The two topologies are *not* the same: the interval $[a, b]$ is open in the Sorgenfrey line but not open in the Euclidean topology (no open interval (c, d) with $c < a < d$ is contained in $[a, b]$). So the Sorgenfrey topology has strictly more open sets than the Euclidean topology, and consequently it also has more closed sets: every Euclidean closed set is closed in $\mathbb{R}_S$.

Some properties of the Sorgenfrey line:

- $[a, +\infty)$ is open, because $[a, +\infty) = \bigcup_{n \in \mathbb{N}} [a, a + n]$ is a union of basic open sets.
- $[a, b]$ is both open and closed (clopen). It is open by definition, and its complement

$$\mathbb{R} \setminus [a, b] = (-\infty, a) \cup [b, +\infty) \quad (1.20)$$

is open, since $(-\infty, a) = \bigcup_{n \in \mathbb{N}} [a - n, a)$ and $[b, +\infty)$ is open as above.

- Closed intervals $[a, b]$ and singletons are closed (they are Euclidean closed), as are all Euclidean closed sets.
- The interval $(a, b]$ is not open in the Sorgenfrey line: the point b belongs to it, but there is no basic set $[b, c)$, $c > b$, contained in $(a, b]$.

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Definition 2.1. Let $X \neq \emptyset$ be a set. A function $d: X \times X \rightarrow \mathbb{R}$ is a **metric** (or **distance function**) on X if for all $x, y, z \in X$:

- (1) $d(x, y) = 0 \iff x = y$;
- (2) $d(x, y) = d(y, x)$;
- (3) $d(x, y) \leq d(x, z) + d(z, y)$ (triangle inequality).

The pair (X, d) is called a *metric space*.

Remark 2.1. A metric is automatically non-negative: for all $x, y \in X$,

$$0 = d(x, x) \leq d(x, y) + d(y, x) = 2d(x, y), \quad \text{so } d(x, y) \geq 0. \quad (2.1)$$

Example 2.1 (Discrete metric)

On any set X , define $d_{\text{dis}}: X \times X \rightarrow \mathbb{R}$ by

$$d_{\text{dis}}(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases} \quad (2.2)$$

This is a function taking only the two values 0 and 1.

Example 2.2 (Euclidean metric)

On $\mathbb{R}$, $d(x, y) = |x - y|$. More generally, on $\mathbb{R}^n$,

$$d(x, y) = \|x - y\| = \left(\sum_{j=1}^n (x_j - y_j)^2 \right)^{\frac{1}{2}}, \quad (2.3)$$

for $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n) \in \mathbb{R}^n$. Properties (1) and (2) are immediate; the triangle inequality follows from the Cauchy–Schwarz inequality for the canonical inner product on $\mathbb{R}^n$. Similarly on $\mathbb{C}^n$,

$$d(x, y) = \|x - y\| = \left(\sum_{j=1}^n |x_j - y_j|^2 \right)^{\frac{1}{2}}, \quad (2.4)$$

where now the metric is induced by the canonical Hermitian product on $\mathbb{C}^n$.

Definition 2.2. Let (X, d) be a metric space, $x \in X$, $r > 0$. The *open ball* centered at x of radius r is

$$B_d(x, r) := \{y \in X \mid d(x, y) < r\} \subseteq X, \quad (2.5)$$

and the *closed ball* is

$$\overline{B}_d(x, r) := \{y \in X \mid d(x, y) \leq r\} \subseteq X. \quad (2.6)$$

Clearly $x \in B_d(x, r) \subseteq \overline{B}_d(x, r)$; in particular the center belongs to the open ball. In $\mathbb{R}$ a ball is an interval, in $\mathbb{R}^2$ a disk, in $\mathbb{R}^3$ a solid ball.

Example 2.3

In a discrete metric space (X, d_{dis}) : if $r > 1$, both $B(x, r)$ and $\overline{B}(x, r)$ equal X , since the metric only takes the values 0, 1. If $r = 1$, then $B(x, 1) = \{x\}$ while $\overline{B}(x, 1) = X$. If $r < 1$, both balls equal $\{x\}$.

Theorem 2.1

Let (X, d) be a metric space. Then

$$\mathcal{B}_d := \{B_d(x, r) \mid x \in X, r > 0\} \quad (2.7)$$

is a base for a topology $\mathcal{T}_d$ on X , called the *metric topology* induced by d .

Proof. We check the two properties of a base from the previous lecture.

(1) $\bigcup_{x,r} B(x, r) = X$, since each x belongs to $B(x, r)$.

(2) Let $B(x_1, r_1)$ and $B(x_2, r_2)$ be two balls, and let $y \in B(x_1, r_1) \cap B(x_2, r_2)$. We must find a ball centered at y contained in the intersection. Set

$$r := \min(r_1 - d(x_1, y), r_2 - d(x_2, y)). \quad (2.8)$$

Both quantities are positive because y lies in both balls, so $r > 0$. We claim $B(y, r) \subseteq B(x_1, r_1) \cap B(x_2, r_2)$. Let $z \in B(y, r)$, so $d(y, z) < r$. By the triangle inequality,

$$d(x_1, z) \leq d(x_1, y) + d(y, z) < d(x_1, y) + r \leq d(x_1, y) + (r_1 - d(x_1, y)) = r_1, \quad (2.9)$$

so $z \in B(x_1, r_1)$. The same argument gives $z \in B(x_2, r_2)$. Since $y \in B(y, r)$, this proves property (2). ■

Remark 2.2. A subset U is open in $\mathcal{T}_d$ if and only if

$$\forall x \in U \quad \exists r > 0 \quad \text{such that } B_d(x, r) \subseteq U. \quad (2.10)$$

That is, around every point of U one can find a sufficiently small ball centered at that point and contained in U .

Definition 2.3. A topological space $(X, \mathcal{T})$ is **metrizable** if there exists a metric d on X which induces the topology of X , i.e. $\mathcal{T} = \mathcal{T}_d$.

So metrizable means that the given topology is induced by some metric, which in general is not known in advance.

Example 2.4

$\mathbb{R}$ with the Euclidean metric is metrizable: the open balls are exactly the bounded open intervals.

The discrete topology X_{dis} is metrizable: for the discrete metric, $B_{d_{\text{dis}}}(x, 1) = \{x\}$ for every x , so all singletons are open, hence every subset is open.

Remark 2.3. The trivial topology X_{triv} is not metrizable if $|X| \geq 2$.

Definition 2.4. The metric topology induced by the Euclidean metric on $\mathbb{R}^n$ (resp. $\mathbb{C}^n$) is the **Euclidean topology**.

This is the *default* topology on $\mathbb{R}^n$ and $\mathbb{C}^n$: when we speak of these as topological spaces without specifying a topology, we always mean the Euclidean one. If another topology is intended, it will always be specified.

Proposition 2.2

Let X be a topological space and $Y \subseteq X$ a subset. Then

$$\mathcal{T}_Y := \{U \cap Y \mid U \subseteq X \text{ open}\} \quad (2.11)$$

is a topology on Y , called the **relative** (or **subspace**) topology.

Thus every subset of a topological space is automatically a topological space with the induced topology. In words:

- $V \subseteq Y$ is open in $Y \iff$ there exists $U \subseteq X$ open with $V = U \cap Y$;
- $C \subseteq Y$ is closed in $Y \iff$ there exists $A \subseteq X$ closed with $C = A \cap Y$ (pass to complements).

Proposition 2.3

Let (X, d) be a metric space and $Y \subseteq X$. Then the restriction $d_Y = d|_{Y \times Y}: Y \times Y \rightarrow \mathbb{R}$ is a metric on Y that induces the relative topology.

Proof. d_Y is clearly a metric (the three axioms hold in particular for points of Y). Moreover, for all $y \in Y$ and $r > 0$,

$$B_{d_Y}(y, r) = B_d(y, r) \cap Y, \quad (2.12)$$

so the balls of d_Y are exactly the traces on Y of the balls of d , and the two topologies coincide. ■

Corollary 2.4

If X is metrizable and $Y \subseteq X$ is a topological subspace, then Y is metrizable.

Remarkable subspaces of $\mathbb{R}^n$

- $B^n := \{x \in \mathbb{R}^n \mid \|x\| \leq 1\} \subseteq \mathbb{R}^n$, the n -ball (closed unit ball);
- $S^n := \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\} \subseteq \mathbb{R}^{n+1}$, the n -sphere;
- $I := [0, 1] \subseteq \mathbb{R}$, the *interval*;
- $\mathbb{R}_+ := [0, +\infty) \subseteq \mathbb{R}$, the *half-line*.

All carry the subspace topology from the Euclidean topology. Note that $S^n \subseteq B^{n+1} \subseteq \mathbb{R}^{n+1}$, and:

- $B^0 = \{0\}$, $B^1 = [-1, 1]$, B^2 is the disk in $\mathbb{R}^2$, B^3 the ball in $\mathbb{R}^3$;
- $S^0 = \{-1, 1\} \subseteq \mathbb{R}$, a discrete space with two points; $S^1 \subseteq \mathbb{R}^2$ is the circle; $S^2 \subseteq \mathbb{R}^3$ is the usual sphere.

The superscript is the *dimension* of the object, not of the ambient space: e.g. the circle S^1 sits in $\mathbb{R}^2$ but is a one-dimensional object.

I , $\mathbb{R}_+$, S^n , B^n are all metrizable, being subspaces of Euclidean space.

Definition 2.5. Let X be a topological space. A subset $J \subseteq X$ is a **neighborhood** of $x \in X$ if there exists an open set $U \subseteq X$ with $x \in U \subseteq J$.

So a neighborhood need not be open. If J is open, it is a neighborhood of each of its points, and is then called an *open neighborhood*.

Remark 2.4. Many textbooks require neighborhoods to be open. Here we use the more flexible definition, because later (compact spaces) we will encounter compact neighborhoods. For local questions, the two notions are essentially equivalent.

Example 2.5

$[-1, 1] \subseteq \mathbb{R}$ is a neighborhood of every $x \in (0, 1)$, but not of -1 or 1 .

Remark 2.5. A subset $U \subseteq X$ is open if and only if for every $x \in U$ there is a neighborhood J of x with $J \subseteq U$. Indeed, if U is open, take $J = U$. Conversely, if the property holds, then every $x \in U$ lies in an open set contained in U ; the union of all these open sets is U , hence U is open.

Definition 2.6. Let $x \in X$. A collection $\mathcal{J}$ of neighborhoods of x is a **base** (or **fundamental system**) of neighborhoods of x if for every neighborhood L of x there exists $J \in \mathcal{J}$ with $J \subseteq L$.

Remark 2.6. It suffices to check this for L an arbitrary open neighborhood of x . Informally: $\mathcal{J}$ contains sufficiently many neighborhoods to understand all the others. This is useful because a point typically has uncountably many neighborhoods, while a base can often be described very simply.

Example 2.6

For $x \in \mathbb{R}$,

$$\mathcal{J}_x = \left\{ \left(x - \frac{1}{n}, x + \frac{1}{n} \right) \mid n \in \mathbb{N}, n \geq 1 \right\} \quad (2.13)$$

is a *countable* base of neighborhoods of x . Indeed, if L is a neighborhood of x , it contains an interval $(x - \varepsilon, x + \varepsilon)$, and choosing n with $\frac{1}{n} < \varepsilon$ gives $(x - \frac{1}{n}, x + \frac{1}{n}) \subseteq L$. The same works at any point of a metric space, using the balls $B(x, \frac{1}{n})$.

Definition 2.7. Let $A \subseteq X$. A subset $J \subseteq X$ is a **neighborhood of A** if there exists an open U with $A \subseteq U \subseteq J$. A collection $\mathcal{J}$ of neighborhoods of A is a **base** (or **fundamental system**) of neighborhoods of A if for every neighborhood L of A there exists $J \in \mathcal{J}$ with $J \subseteq L$.

Let X be a topological space and $A \subseteq X$.

Definition 2.8 (Interior). The **interior** of A in X is

$$\text{Int}_X A := \bigcup_{\substack{U \subseteq A \\ U \text{ open}}} U, \quad (2.14)$$

the union of all open subsets of X contained in A .

It is sometimes denoted by $\overset{\circ}{A}$, but this notation hides the fact that it depends on the ambient space X , i.e. on how A sits inside X .

- Remark 2.7.**
1. $\text{Int}_X A$ is the largest open subset of X contained in A (possibly empty).
 2. $\text{Int}_X A \subseteq A$, with equality if and only if A is open in X .
 3. If $U \subseteq A$ and U is open in X , then $U \subseteq \text{Int}_X A$.
 4. $x \in \text{Int}_X A$ if and only if there is a neighborhood U of x with $U \subseteq A$, i.e. if and only if A is a neighborhood of x .

Example 2.7

In the Euclidean topology,

$$\text{Int}_{\mathbb{R}} [0, 1] = (0, 1), \quad \text{Int}_{\mathbb{R}} \{0\} = \emptyset. \quad (2.15)$$

In the Sorgenfrey line (the topology on $\mathbb{R}$ with basis the half-open intervals $[a, b)$), $\text{Int} [0, 1] = [0, 1)$, since $[0, 1)$ is a basic open set, whereas it is not open in the Euclidean topology.

Definition 2.9 (Exterior). $\text{Ext}_X A := \text{Int}_X (X \setminus A)$.

It is the largest open subset of X not intersecting A , and it has properties analogous to those of the interior.

Definition 2.10 (Closure). The **closure** of A in X is

$$\text{Cl}_X A := \bigcap_{\substack{C \supseteq A \\ C \text{ closed}}} C, \quad (2.16)$$

the intersection of all closed subsets of X containing A .

It is also written $\overline{A}$, but again the notation Cl_X records the dependence on the ambient space.

Remark 2.8. 1. $\text{Cl}_X A$ is the smallest closed subset of X containing A , being an intersection of closed sets.

2. $A \subseteq \text{Cl}_X A$, with equality if and only if A is closed.

3. If $A \subseteq C$ and C is closed in X , then $\text{Cl}_X A \subseteq C$.

Proposition 2.5 (Membership criterion for the closure)

$x \in \text{Cl}_X A$ if and only if for every (open) neighborhood U of x in X , we have $U \cap A \neq \emptyset$.

Intuitively, a point of the closure of A must be “very close” to A . Restricting to open neighborhoods is harmless, since every neighborhood contains an open one.

Proof. ($\Rightarrow$) Let $x \in \text{Cl}_X A$, and suppose by contradiction that some open neighborhood U of x satisfies $U \cap A = \emptyset$. Then $A \subseteq X \setminus U$, which is closed because U is open. By minimality of the closure, $\text{Cl}_X A \subseteq X \setminus U$. But $x \in \text{Cl}_X A$ and $x \in U$, a contradiction.

($\Leftarrow$) Suppose every open neighborhood of x meets A , and suppose by contradiction that $x \notin \text{Cl}_X A$. Then $U := X \setminus \text{Cl}_X A$ is open (the closure is closed) and contains x . Since $A \subseteq \text{Cl}_X A$,

$$U \cap A \subseteq U \cap \text{Cl}_X A = \emptyset, \quad (2.17)$$

so U is an open neighborhood of x not meeting A , contradicting the hypothesis. ■

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Let X be a topological space and $A \subset X$.

Membership criteria. By the criterion for the closure (last lecture), there are analogous criteria for the other operators:

- $x \in \text{Int}_X A$ if and only if there exists a neighborhood U of x with $U \subset A$;
- $x \in \text{Ext}_X A$ if and only if there exists a neighborhood U of x with $U \cap A = \emptyset$;
- $x \in \text{Cl}_X A$ if and only if every (open) neighborhood U of x satisfies $U \cap A \neq \emptyset$.

Definition 3.1 (Frontier). The **frontier** (or **boundary**) of A in X is

$$\text{Fr}_X A = \partial_X A := \text{Cl}_X A \cap \text{Cl}_X (X \setminus A). \quad (3.1)$$

Remark. $\text{Fr}_X A$ is closed in X (an intersection of two closed sets) and $\text{Fr}_X A \subset \text{Cl}_X A$. Moreover, by the membership criterion for the closure applied to A and to $X \setminus A$,

$$x \in \text{Fr}_X A \iff \text{every neighborhood } U \text{ of } x \text{ satisfies } U \cap A \neq \emptyset \text{ and } U \cap (X \setminus A) \neq \emptyset. \quad (3.2)$$

So the point x is “on the boundary”: arbitrarily close to both A and its complement.

Theorem 3.1

The following equalities hold:

- (1) $\text{Cl}_X A = X \setminus \text{Ext}_X A$;
- (2) $\text{Fr}_X A = \text{Cl}_X A \setminus \text{Int}_X A$.

Proof. (1) $(\subset)$ $\text{Ext}_X A$ is open, so $X \setminus \text{Ext}_X A$ is closed. Also $\text{Ext}_X A \subset X \setminus A$, hence $A \subset X \setminus \text{Ext}_X A$. Since $\text{Cl}_X A$ is the smallest closed set containing A , we get $\text{Cl}_X A \subset X \setminus \text{Ext}_X A$.

$(\supset)$ $X \setminus \text{Cl}_X A$ is open and, since $A \subset \text{Cl}_X A$, it is contained in $X \setminus A$. Since $\text{Ext}_X A = \text{Int}_X (X \setminus A)$ is the largest open subset of $X \setminus A$, we get $X \setminus \text{Cl}_X A \subset \text{Ext}_X A$. Passing to complements, $X \setminus \text{Ext}_X A \subset \text{Cl}_X A$.

(2) Apply (1) to the subset $X \setminus A$:

$$\text{Cl}_X (X \setminus A) = X \setminus \text{Ext}_X (X \setminus A) = X \setminus \text{Int}_X (X \setminus (X \setminus A)) = X \setminus \text{Int}_X A. \quad (3.3)$$

Therefore

$$\text{Fr}_X A = \text{Cl}_X A \cap \text{Cl}_X (X \setminus A) = \text{Cl}_X A \cap (X \setminus \text{Int}_X A) = \text{Cl}_X A \setminus \text{Int}_X A. \quad (3.4)$$

■

Consequence. For any $A \subset X$, the space X decomposes as a *disjoint* union

$$X = \text{Int}_X A \sqcup \text{Fr}_X A \sqcup \text{Ext}_X A. \quad (3.5)$$

Indeed $\text{Cl}_X A = \text{Int}_X A \sqcup \text{Fr}_X A$ by (2), and $X = \text{Cl}_X A \sqcup \text{Ext}_X A$ by (1).

Remark. From (2):

- A is both open and closed if and only if $\text{Fr}_X A = \emptyset$ (then $\text{Cl}_X A = \text{Int}_X A$, which forces A to coincide with both).
- If A is closed with empty interior, then $\text{Fr}_X A = A$.

Examples.

1. In $\mathbb{R}$ with the Sorgenfrey topology (basis $[a, b)$), consider $A = [0, +\infty)$. It is open, being a union of basic sets. Its complement $(-\infty, 0)$ is also open in the Sorgenfrey topology, so A is closed as well. Hence A is clopen and $\text{Fr } A = \emptyset$. (In the Euclidean topology, $\text{Fr } A = \{0\}$.) Recall that the Euclidean topology is coarser than the Sorgenfrey topology.
2. $\text{Int}_{\mathbb{R}} \mathbb{Q} = \emptyset$. Every nonempty open interval contains rational numbers, so by the membership criterion every real number belongs to $\text{Cl}_{\mathbb{R}} \mathbb{Q}$, i.e. $\mathbb{Q}$ is *dense* in $\mathbb{R}$ and $\text{Cl}_{\mathbb{R}} \mathbb{Q} = \mathbb{R}$. Likewise every open interval contains irrational numbers, so $\mathbb{R} \setminus \mathbb{Q}$ is dense too. Hence

$$\text{Fr}_{\mathbb{R}} \mathbb{Q} = \text{Cl}_{\mathbb{R}} \mathbb{Q} \cap \text{Cl}_{\mathbb{R}} (\mathbb{R} \setminus \mathbb{Q}) = \mathbb{R}, \quad (3.6)$$

and $\text{Ext}_{\mathbb{R}} \mathbb{Q} = \emptyset$.

Definition 3.2. A subset $A \subset X$ is *dense* in X if $\text{Cl}_X A = X$.

Definition 3.3. Let X and Y be topological spaces. A map $f: X \rightarrow Y$ is *continuous* if

$$\forall V \subset Y \text{ open} \implies f^{-1}(V) \subset X \text{ open}. \quad (3.7)$$

That is, preimages of open sets of the codomain are open in the domain. This is a purely topological definition. The ε - δ definition seen in calculus is an *operational* characterization that is convenient for proving that a given function is continuous, in the special case of Euclidean topologies; but to develop a general theory, the topological definition is the right one.

Remark. For every map and every subset $V \subset Y$, we have $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$. Hence

$$f \text{ is continuous} \iff \forall C \subset Y \text{ closed} \implies f^{-1}(C) \subset X \text{ closed}. \quad (3.8)$$

Proof of the identity. Let $x \in X$. Then $x \in f^{-1}(Y \setminus V)$ iff $f(x) \in Y \setminus V$ iff $f(x) \notin V$ iff $x \notin f^{-1}(V)$ iff $x \in X \setminus f^{-1}(V)$. All the steps are reversible. ■

Proposition 3.2

If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are continuous maps between topological spaces, then $g \circ f: X \rightarrow Z$ is continuous.

Proof. Let $V \subset Z$ be open. Then

$$(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V)). \quad (3.9)$$

Since g is continuous, $g^{-1}(V)$ is open in Y ; since f is continuous, $f^{-1}(g^{-1}(V))$ is open in X . ■

Remarks (basic examples of continuous maps).

1. The identity $\text{id}_X: X \rightarrow X$ is continuous (with the same topology on domain and codomain).
2. Every constant map $c: X \rightarrow Y$ is continuous: if $V \subset Y$ is open and contains the value y_0 , then $c^{-1}(V) = X$, which is open; otherwise $c^{-1}(V) = \emptyset$, also open.
3. If $Y \subset X$ is a topological subspace, the inclusion $i_Y: Y \hookrightarrow X$ is continuous, since $i_Y^{-1}(U) = U \cap Y$, which is open in Y by definition of the relative topology.
4. Restrictions of continuous maps are continuous: $f|_Y = f \circ i_Y$ is a composition of continuous maps.
5. Every map $f: X_{\text{dis}} \rightarrow Y$ is continuous, since every subset of X_{dis} is open.
6. Every map $f: X \rightarrow Y_{\text{triv}}$ is continuous: the only open sets of Y_{triv} are $\emptyset$ and Y , with preimages $\emptyset$ and X , both open.

Remark (a converse to 5). If every map from X to any topological space is continuous, then X is discrete. Indeed, consider $\text{id}: X \rightarrow X_{\text{dis}}$; it is continuous by hypothesis, so for every subset $S \subset X$, the set $\text{id}^{-1}(S) = S$ is open in X . (In categorical language, the empty space is the initial object and the one-point space is the terminal object in the category of topological spaces with continuous maps; but note that “there is a continuous map to Y for every Y ” is different from “every map is continuous”. In general between two spaces there are many continuous maps, because continuity is a rather flexible requirement.)

Open and closed maps

Definition 3.4. A map $f: X \rightarrow Y$ is

- (1) *open* if for every $U \subset X$ open, $f(U)$ is open in Y ;
- (2) *closed* if for every $C \subset X$ closed, $f(C)$ is closed in Y .

These definitions do *not* assume continuity. However, the key notion is continuity: whenever we consider maps between topological spaces we always assume they are continuous, possibly with the additional requirement of being open, closed, or both.

Remark. A constant map $c: \mathbb{R} \rightarrow \mathbb{R}$ is continuous and closed (its image is a point, which is closed), but *not* open (a point is not open in $\mathbb{R}$). So continuity does not imply openness. Also, for $A \subset X$: A is open (resp. closed) in X if and only if the inclusion $i_A: A \hookrightarrow X$ is an open (resp. closed) map.

Example (a bijective continuous map which is not open). Let $\mathbb{R}_\ell$ denote $\mathbb{R}$ with the Sorgenfrey topology (or, as in the notes, $\mathbb{R}_{\text{dis}}$ the discrete line). The identity

$$\text{id}_{\mathbb{R}}: \mathbb{R}_\ell \rightarrow \mathbb{R} \tag{3.10}$$

is continuous, because every Euclidean open set is open in the finer topology, and it is a bijection. But it is not open: the image of the basic open set $[0, 1)$ is $[0, 1)$, which is not open in the Euclidean topology. Equivalently, the inverse map $\mathbb{R} \rightarrow \mathbb{R}_\ell$ is not continuous. The same works in general whenever domain and codomain have the same underlying set and the domain topology is strictly finer than the codomain topology; in particular $\text{id}_{\mathbb{R}}: \mathbb{R}_{\text{dis}} \rightarrow \mathbb{R}$ is a continuous bijection with discontinuous inverse.

Contrast this with algebra: a bijective group homomorphism has an inverse which is automatically a homomorphism. In topology a continuous bijection need *not* have a continuous inverse. This leads to the right notion of isomorphism.

Definition 3.5. A map $f: X \rightarrow Y$ is a *homeomorphism* if

- (1) f is bijective;
- (2) f is continuous;
- (3) f^{-1} is continuous.

X and Y are *homeomorphic* if there exists a homeomorphism $f: X \rightarrow Y$, and we write $X \cong Y$.

Remarks.

- id_X is a homeomorphism.
- If $f: X \rightarrow Y$ is a homeomorphism, so is $f^{-1}: Y \rightarrow X$.
- If $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ are homeomorphisms, so is $g \circ f$.

Hence being homeomorphic is an “equivalence relation” between topological spaces.

Remark. If $f: X \rightarrow Y$ is bijective, then

$$f^{-1} \text{ continuous} \iff f \text{ open} \iff f \text{ closed.} \tag{3.11}$$

Indeed, for a bijection, $(f^{-1})^{-1}(U) = f(U)$, so continuity of f^{-1} says exactly that f maps open sets to open sets; passing to complements gives the closed version. Consequently

$$f \text{ homeomorphism} \iff f \text{ continuous, bijective and open (or closed)}. \quad (3.12)$$

This is the only situation in which openness and continuity of the inverse coincide: it requires f to be bijective.

Corollary. The set

$$\text{Homeo}(X) := \{f: X \rightarrow X \mid f \text{ homeomorphism}\} \quad (3.13)$$

is a group with respect to composition (the *homeomorphism group* of X): the identity is the neutral element, compositions of homeomorphisms are homeomorphisms, and inverses exist. For example, $\text{Homeo}(\mathbb{R})$ is a very large group, which one cannot describe explicitly: it contains every strictly increasing or strictly decreasing continuous bijection of $\mathbb{R}$, including very complicated ones, e.g. nowhere differentiable ones.

Definition 3.6. A property $\mathcal{P}$ is a *topological property* if for all topological spaces X, Y : if X has $\mathcal{P}$ and $X \cong Y$, then Y has $\mathcal{P}$.

That is, a topological property is invariant under homeomorphisms; topology studies the topological properties of topological spaces. For instance, connectedness is a topological property, as we will see later in the course.

Proposition 3.3

Metrizability is a topological property.

Proof. Let X be metrizable, with a metric d_X inducing the topology of X , and let Y be a space with $X \cong Y$. Take a homeomorphism $f: Y \rightarrow X$ (the inverse of a homeomorphism from X to Y). Define

$$d_Y: Y \times Y \rightarrow \mathbb{R}, \quad d_Y(y_1, y_2) = d_X(f(y_1), f(y_2)). \quad (3.14)$$

This is a metric on Y : positivity and symmetry are inherited from d_X , $d_Y(y_1, y_2) = 0$ iff $f(y_1) = f(y_2)$ iff $y_1 = y_2$ by injectivity of f , and the triangle inequality follows from the one for d_X . Since f is a homeomorphism, f maps d_Y -balls to d_X -balls: $B_{d_Y}(y, r) = f^{-1}(B_{d_X}(f(y), r))$ (using surjectivity of f). Hence the topology induced by d_Y is exactly the topology of Y (exercise: write out this verification using the definition of the metric topology). ■

Definition 3.7. $C(X, Y) := \{f: X \rightarrow Y \mid f \text{ continuous}\}$.

Remark. $C(X, Y) \neq \emptyset$ (there are constant maps), and $\text{Homeo}(X) \subset C(X, X)$.

Examples of homeomorphic spaces

Example 1. The map

$$\exp: \mathbb{R} \rightarrow (0, +\infty), \quad x \mapsto e^x \quad (3.15)$$

is continuous and bijective with continuous inverse $\log$. Hence $\mathbb{R} \cong (0, +\infty)$, an unbounded interval.

Example 2. The map

$$(0, 1) \rightarrow (0, +\infty), \quad x \mapsto \frac{x}{1-x} \quad (3.16)$$

is continuous and bijective; its inverse $y \mapsto \frac{y}{1+y}$ is a rational function, hence continuous. So $(0, 1) \cong (0, +\infty)$.

Since homeomorphism is an equivalence relation, combining the two examples,

$$\mathbb{R} \cong (0, +\infty) \cong (0, 1), \quad (3.17)$$

i.e. $\mathbb{R}$ is homeomorphic to every open interval, bounded or not.

Remark. In particular *being bounded is not a topological property*. (Boundedness only makes sense in a metric space; every metric space admits a bounded metric inducing the same topology, but it is false that if a space admits one bounded metric then all its metrics are bounded.)

Remark (continuity at a point). In this course, “continuous” always means continuous at every point, i.e. globally continuous in the sense of the definition above; we do not consider continuity at a single point.